Lab Sheet 4: SQL Operators

Arithmetic, Logical, Comparison, Special and Set Operators - SQL Answers with Expected Outputs (MySQL 8.x)

Name: _____ Roll No: _____ Section: _____

Database and sample data are the ones given in the lab sheet. Question 22 is not present in the lab sheet, so numbering goes from 21 to 23.

Setup (run once before Question 1)

```
CREATE DATABASE Lab4_Operators;
USE Lab4_Operators;

CREATE TABLE Product (ProductID INT PRIMARY KEY AUTO_INCREMENT, ProductName VARCHAR(50),
                      Price DECIMAL(10,2), Quantity INT);
CREATE TABLE Student (StudentID INT PRIMARY KEY AUTO_INCREMENT, Name VARCHAR(50),
                      Course VARCHAR(50), City VARCHAR(50), Fees DECIMAL(10,2));
CREATE TABLE Employee (EmpID INT PRIMARY KEY AUTO_INCREMENT, Name VARCHAR(50),
                       Department VARCHAR(50), Salary DECIMAL(10,2), City VARCHAR(50));

INSERT INTO Product (ProductName, Price, Quantity) VALUES
('Laptop', 65000.00, 2), ('Mouse', 1500.00, 5), ('Keyboard', 2500.00, 3), ('Monitor', 12000.00, 1);
INSERT INTO Student (Name, Course, City, Fees) VALUES
('Amit Sharma', 'B.Tech CSE', 'Roorkee', 75000.00), ('Anjali Rai', 'B.Tech CSE', 'Delhi', 80000.00),
('Akash Verma', 'B.Tech ECE', 'Roorkee', 60000.00), ('Deepak Kumar', 'B.Com', NULL, 45000.00);
INSERT INTO Employee (Name, Department, Salary, City) VALUES
('Rahul Dev', 'IT', 58000.00, 'Delhi'), ('Sonia Malhotra', 'HR', 52000.00, 'Roorkee'),
('Vikas Singh', 'IT', 70000.00, 'Delhi'), ('Anand Mehta', 'Finance', 65000.00, 'Delhi');
```

Arithmetic Operators

1. Perform addition of two numbers using +.

SQL Answer:

```
SELECT 25 + 15 AS AdditionResult;
```

Expected Output:

AdditionResult
40

2. Perform subtraction using -.

SQL Answer:

```
SELECT 50 - 18 AS SubtractionResult;
```

Expected Output:

SubtractionResult
32

3. Perform multiplication using *.

SQL Answer:

```
SELECT 12 * 8 AS MultiplicationResult;
```

Expected Output:

MultiplicationResult
96

4. Perform division using /.

SQL Answer:

```
SELECT 100 / 4 AS DivisionResult;
```

Expected Output:

DivisionResult
25.0000

5. Calculate the total amount as Price * Quantity.

SQL Answer:

```
SELECT ProductName, Price, Quantity,
       (Price * Quantity) AS TotalAmount
FROM Product;
```

Expected Output:

ProductName	Price	Quantity	TotalAmount
Laptop	65000.00	2	130000.00
Mouse	1500.00	5	7500.00
Keyboard	2500.00	3	7500.00
Monitor	12000.00	1	12000.00

Logical Operators

6. Demonstrate AND with two conditions.

SQL Answer:

```
SELECT * FROM Student
WHERE City = 'Delhi' AND Fees > 65000;
```

Expected Output:

StudentID	Name	Course	City	Fees
2	Anjali Rai	B.Tech CSE	Delhi	80000.00

7. Demonstrate OR with two conditions.

SQL Answer:

```
SELECT * FROM Student
WHERE City = 'Delhi' OR City = 'Roorkee';
```

Expected Output:

StudentID	Name	Course	City	Fees
1	Amit Sharma	B.Tech CSE	Roorkee	75000.00
2	Anjali Rai	B.Tech CSE	Delhi	80000.00
3	Akash Verma	B.Tech ECE	Roorkee	60000.00

8. Demonstrate NOT with a condition.

SQL Answer:

```
SELECT * FROM Student
WHERE NOT (City = 'Delhi');
```

Expected Output:

StudentID	Name	Course	City	Fees
1	Amit Sharma	B.Tech CSE	Roorkee	75000.00
3	Akash Verma	B.Tech ECE	Roorkee	60000.00

Note: Deepak (City is NULL) is not shown, because NOT (NULL = Delhi) is NULL, not TRUE.

9. Find students belonging to a specific course AND city.

SQL Answer:

```
SELECT * FROM Student
WHERE Course = 'B.Tech CSE' AND City = 'Roorkee';
```

Expected Output:

StudentID	Name	Course	City	Fees
1	Amit Sharma	B.Tech CSE	Roorkee	75000.00

10. Find employees belonging to either of two departments.

SQL Answer:

```
SELECT * FROM Employee
WHERE Department = 'IT' OR Department = 'HR';
```

Expected Output:

EmpID	Name	Department	Salary	City
1	Rahul Dev	IT	58000.00	Delhi
2	Sonia Malhotra	HR	52000.00	Roorkee
3	Vikas Singh	IT	70000.00	Delhi

Comparison Operators

11. Demonstrate the = operator.

SQL Answer:

```
SELECT * FROM Student WHERE City = 'Delhi';
```

Expected Output:

StudentID	Name	Course	City	Fees
2	Anjali Rai	B.Tech CSE	Delhi	80000.00

12. Demonstrate the > operator.

SQL Answer:

```
SELECT * FROM Employee WHERE Salary > 55000.00;
```

Expected Output:

EmpID	Name	Department	Salary	City
1	Rahul Dev	IT	58000.00	Delhi
3	Vikas Singh	IT	70000.00	Delhi
4	Anand Mehta	Finance	65000.00	Delhi

13. Demonstrate the < operator.

SQL Answer:

```
SELECT * FROM Product WHERE Price < 2000.00;
```

Expected Output:

ProductID	ProductName	Price	Quantity
2	Mouse	1500.00	5

14. Demonstrate >= and <=.

SQL Answer:

```
SELECT * FROM Employee
WHERE Salary >= 50000.00 AND Salary <= 65000.00;
```

Expected Output:

EmpID	Name	Department	Salary	City
1	Rahul Dev	IT	58000.00	Delhi
2	Sonia Malhotra	HR	52000.00	Roorkee
4	Anand Mehta	Finance	65000.00	Delhi

15. Demonstrate the <> / != operator.

SQL Answer:

```
SELECT * FROM Student WHERE City != 'Delhi';
```

Expected Output:

StudentID	Name	Course	City	Fees
1	Amit Sharma	B.Tech CSE	Roorkee	75000.00
3	Akash Verma	B.Tech ECE	Roorkee	60000.00

Special Operators

16. Demonstrate BETWEEN for a salary range.

SQL Answer:

```
SELECT * FROM Employee
WHERE Salary BETWEEN 50000.00 AND 65000.00;
```

Expected Output:

EmpID	Name	Department	Salary	City
1	Rahul Dev	IT	58000.00	Delhi
2	Sonia Malhotra	HR	52000.00	Roorkee
4	Anand Mehta	Finance	65000.00	Delhi

Note: BETWEEN is inclusive of both ends.

17. Demonstrate IN for selecting multiple cities.

SQL Answer:

```
SELECT * FROM Student
WHERE City IN ('Delhi', 'Roorkee', 'Dehradun');
```

Expected Output:

StudentID	Name	Course	City	Fees
1	Amit Sharma	B.Tech CSE	Roorkee	75000.00
2	Anjali Rai	B.Tech CSE	Delhi	80000.00
3	Akash Verma	B.Tech ECE	Roorkee	60000.00

18. Demonstrate NOT IN.

SQL Answer:

```
SELECT * FROM Student
WHERE City NOT IN ('Delhi', 'Meerut');
```

Expected Output:

StudentID	Name	Course	City	Fees
1	Amit Sharma	B.Tech CSE	Roorkee	75000.00
3	Akash Verma	B.Tech ECE	Roorkee	60000.00

19. Demonstrate LIKE for names beginning with a particular letter.

SQL Answer:

```
SELECT * FROM Student WHERE Name LIKE 'A%';
```

Expected Output:

StudentID	Name	Course	City	Fees
1	Amit Sharma	B.Tech CSE	Roorkee	75000.00
2	Anjali Rai	B.Tech CSE	Delhi	80000.00
3	Akash Verma	B.Tech ECE	Roorkee	60000.00

20. Demonstrate IS NULL and IS NOT NULL.

SQL Answer:

```
SELECT * FROM Student WHERE City IS NULL;
SELECT * FROM Student WHERE City IS NOT NULL;
```

Expected Output:

IS NULL:

StudentID	Name	Course	City	Fees
4	Deepak Kumar	B.Com	NULL	45000.00

IS NOT NULL:

StudentID	Name	Course	City	Fees
1	Amit Sharma	B.Tech CSE	Roorkee	75000.00
2	Anjali Rai	B.Tech CSE	Delhi	80000.00
3	Akash Verma	B.Tech ECE	Roorkee	60000.00

Set Operations and Integrated Query

21. Demonstrate UNION using two compatible queries.

SQL Answer:

```
SELECT Name, City FROM Student
UNION
SELECT Name, City FROM Employee;
```

Expected Output:

Name	City
Amit Sharma	Roorkee
Anjali Rai	Delhi
Akash Verma	Roorkee
Deepak Kumar	NULL
Rahul Dev	Delhi
Sonia Malhotra	Roorkee
Vikas Singh	Delhi
Anand Mehta	Delhi

Note: UNION removes duplicate rows. Use UNION ALL to keep them.

23. Demonstrate INTERSECT where supported by the DBMS.

SQL Answer:

```
SELECT City FROM Student
INTERSECT
SELECT City FROM Employee;
```

Expected Output:

City
Roorkee
Delhi

Note: INTERSECT works in MySQL 8.0.31 and later.

24. Demonstrate EXCEPT / MINUS where supported.

SQL Answer:

```
SELECT City FROM Student
EXCEPT
SELECT City FROM Employee;
```

Expected Output:

City
NULL

Note: MySQL supports EXCEPT (8.0.31 and later), not MINUS.

25. Perform a problem using Arithmetic + Logical + Comparison + Special operators.

SQL Answer:

```
SELECT EmpID, Name, Salary, (Salary * 12) AS AnnualSalary
FROM Employee
WHERE (Salary * 12) > 600000
  AND Department IN ('IT', 'Finance')
  AND City = 'Delhi'
  AND Salary IS NOT NULL;
```

Expected Output:

EmpID	Name	Salary	AnnualSalary
1	Rahul Dev	58000.00	696000.00
3	Vikas Singh	70000.00	840000.00
4	Anand Mehta	65000.00	780000.00